W1 · Discovering questions worth investigating

Business Intelligence IIB423T-1 · Universidad del Desarrollo · Instructor: Tomás Fontecilla Correa

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Summary. This W1 checkpoint brings two candidate lines of inquiry, both based on FONASA's public Diagnosis-Related Groups (GRD) files for discharge years 2023 and 2024 (1,039,587 and 1,085,813 episodes). **A** examines hospitalisation outside the patient's region of residence within the network of GRD-funded hospitals. **B** examines case-mix-adjusted length of stay and the weight of extreme stays by hospital. Real data were inspected for both. A shows clear regional differences (Metropolitan region ≈ 0.9% vs O'Higgins ≈ 6.0% in 2024), but its denominator is not population-based. B shows that extreme stays (≈ 6.3% of episodes) account for ≈ 25% of bed-days, and that an adjusted stay index (IEMA) is stable between years (r = 0.93). Our **provisional** preference is **B**. This is an initial framing only: full cleaning, the complete EDA and the final question are left for C1, after instructor feedback (section 6).

Starting point. We used the course repository (*Mxtiilxz/Workshop*: source inventory and download script) and, as a methodological reference, the public repository *thenoeyedgirl/proyecto-GRD-Fonasa* (another team's regional GRD analysis for 2023-2024). We adopted its territorial focus as candidate A, tested its assumptions and corrected three of them (section 3.3). Candidate B is our own.

1. Candidate A · Hospitalisation outside the region of residence

Problem. In Chile, high-complexity care and medical specialists are concentrated in a few regions. Some patients are admitted to a hospital in another region, with travel costs for families and pressure on receiving centres. Population: inpatient stays (excluding major ambulatory surgery, CMA) in the public hospitals funded through GRD, 2023-2024, by region of residence.

Documented relevance. A report by the Departamento de Estudios y Desarrollo (July 2023) states that the Metropolitan region concentrates 59.7% of the country's certified medical specialists (20,846 of 34,944, September 2022). Tarapacá has 0.70 specialists per 1,000 inhabitants against a national 1.76 [4]. FONASA describes GRD as a payment mechanism for a network of public hospitals (68 in 2023) and contracted private clinics [3]. Policy link: MINSAL's National Health Strategy for the 2030 Health Objectives (2022 edition) includes Strategic Axis 7, «Management, quality and innovation» [5]. We did not verify an indicator or target in that strategy that directly measures inter-regional referral, so we do *not* equate our indicator with a health target.

Tentative question. What share of inpatient stays of residents of each region takes place in GRD hospitals of another region, where do they go, and for which kinds of cases (GRD weight, elective or emergency admission)? Did it change between 2023 and 2024?

Proposed user and use (our proposal). The Undersecretariat of Healthcare Networks and the sending Health Services could use it to prioritise regional capacity (beds or specialties) and to size referral agreements and travel support.

Data opportunity. The GRD files record the municipality of residence, the hospital, the GRD, its weight, admission type and origin. Missing: stays in non-GRD facilities (lower-complexity hospitals, private clinics, armed forces) and the reference population. A population denominator would need DEIS discharges from all facilities or INE population (2024 Census).

2. Candidate B · Case-mix-adjusted length of stay and extreme stays

Problem. Length of stay (LOS) drives bed use, the critical resource of public hospitals. Comparing crude LOS between hospitals is unfair because each hospital treats a different case mix; GRD allows that case mix to be adjusted for. Population: inpatient stays in GRD hospitals, 2023-2024.

Documented relevance. Zapata (2018) explains that GRD groups clinically and economically similar episodes and that it allows case-mix-adjusted LOS to be compared. The author reports a 2014 national norm with a mean LOS of 6.2 days and 6.4% upper outliers over 807,195 discharges [6]. FONASA (2023) reports that in 2023 it allocated resources through GRD to 68 facilities and that hospital complexity remained high after the pandemic [3]. Policy link: Axis 7 «Management, quality and innovation» of the 2022 National Strategy [5]. Here too we found no quantitative LOS target that we can cite.

Tentative question. Which hospitals have a systematically longer stay than expected for their case mix (IEMA > 1)? How much of that excess comes from extreme stays? Is the signal stable between 2023 and 2024?

Proposed user and use (our proposal). FONASA (Financing Models Department) and hospital bed-management teams could use it to target reviews of prolonged stays and complex discharges in the GRDs and hospitals where excess bed-days concentrate.

Data opportunity. The files contain admission and discharge dates, the GRD (IR 3.0), severity, mortality, discharge type, internal transfers and specialty. Missing: the official LOS norm per GRD (we use an internal benchmark), bed-waiting time, social reasons for delayed discharge and cost per day.

3. Inspection of real data

3.1 Source, version and unit of observation

Source. FONASA, «Grupos Relacionados por el Diagnóstico (GRD)» dashboard, Data Catalogue tab [1]: GRD_PUBLICO_2023.zip (extracted by FONASA on 04-09-2024) and GRD_PUBLICO_2024.zip (extracted on 16-05-2025), downloaded on 25-09-2026. The catalogue warns that each file is a «snapshot» of what was coded at extraction time. Dictionary: «Tablas maestras bases GRD» [2]. For hospital location and complexity we used the current health-facilities file (MINSAL, datos.gob.cl) included in the reference repository [8]. SHA-256 fingerprints of every file are recorded in the README.

What a row is. One episode coded in GRD, i.e. an inpatient discharge or a major ambulatory surgery (CMA), from one of the GRD-funded hospitals. It is not a patient or a bed-day. Files are organised by **discharge year**: every discharge in the 2023 file falls in 2023, while admission may be earlier. Fields used: hospital (DEIS code), municipality and province of residence, episode type, admission and discharge dates, admission type, origin, discharge type, IR-GRD code (6 digits), weight, severity and mortality (0 to 3). An encrypted patient identifier exists but was left out of the extract because W1 does not need it.

Discharge year	Episodes	Inpatient stays	CMA	Hospitals	Municipalities of residence
2023	1,039,587	846,043	193,544	68	347
2024	1,085,813	873,772	212,041	72	344

Table 1. Coverage by year. Four hospitals join in 2024 (Constitución, Cauquenes, Penco-Lirquén and Padre Las Casas). Comparisons between years use only the 68 common hospitals.

3.2 First quality checks (Five Cs)

We applied an operational reading of the Five Cs (completeness, consistency, conformity, coverage and credibility). If the Class 03 definition differs, the checks are re-labelled but their content stays the same.

C	Check	Result
Completeness	FECHA_INGRESO (admission date) not parseable	52 (0.0024%)
Completeness	FECHAALTA (discharge date) not parseable	0 (0.0000%)
Completeness	COMUNA = 'DESCONOCIDO' (unknown) or empty	12 (0.0006%)
Completeness	Age not computable	65 (0.0031%)
Completeness	HOSPPROCEDENCIA empty (expected: only transfers have it)	1,892,677 (89.0504%)
Consistency	Negative length of stay (discharge < admission)	0 (0.0000%)
Consistency	CMA with length of stay > 1 day	419 (0.0197%)
Consistency	Age < 0 or > 110	0 (0.0000%)
Conformity	GRD code not in the IR-GRD dictionary	47 (0.0022%)
Conformity	GRD weight not numeric	45 (0.0021%)
Conformity	Severity outside {0,1,2,3}	45 (0.0021%)
Coverage	Hospitals 2023 / 2024	68 / 72
Credibility	FONASA extraction date 2023 / 2024	04-09-2024 / 16-05-2025 (FONASA catalogue)
Consistency	Different schema between years	2023: UTF-16, CIP_ENCRIPTADO, dates DD-MM-YYYY; 2024: Latin-1, ID_BENEFICIARIO, dates YYYY-MM-DD
Consistency	Region labels in the facilities file	«De» and «de» mixed; 2 municipalities assigned to 2 regions (Longaví, Tucapel), resolved by mode

Table 2. Initial checks on 2,125,400 episodes. Critical fields are almost complete. The relevant problems concern consistency between versions and coverage, not missing data.

3.3 Checking the reference repository

- **Dates.** The reference reads dates as DD/MM/YYYY for both years. In the data, 0% of dates match that format: 2023 uses DD-MM-YYYY and 2024 uses YYYY-MM-DD. Its notebook reports only 211,157 valid stays with a mean of 110.9 days. Recomputing with each year's format gives 1.72 million valid stays with a mean of ≈ 7 days, consistent with the 2014 norm of 6.2 days [6].
- **SERVICIO_SALUD.** The reference uses it as the hospital's location. However, each hospital records a median of 18.5 different health services (up to 30), so the field describes the *patient*. We locate hospitals through the DEIS code joined to the facilities file.
- **Non-local by municipality (48.6%) and expected region by modal province.** A high-complexity hospital treating patients from other municipalities is expected by network design, and a region assigned by mode is circular. We use the region of residence derived from the municipality. We also exclude CMA episodes (19.1%), which the reference mixes with inpatient stays.

3.4 Candidate A: first observations

Joins. (1) COD_HOSPITAL → facility code: **verified**, 72 of 72 hospitals match (58 high and 14 medium complexity). (2) Municipality of residence (text) → region, through a municipality-region table derived from the facilities file with normalised labels: **proposed and tested**, 12 episodes (0.0006%) remain without a region. Matching names do not guarantee the same meaning: the hospital's region comes from the current version of the facilities file, not necessarily the 2023 one.

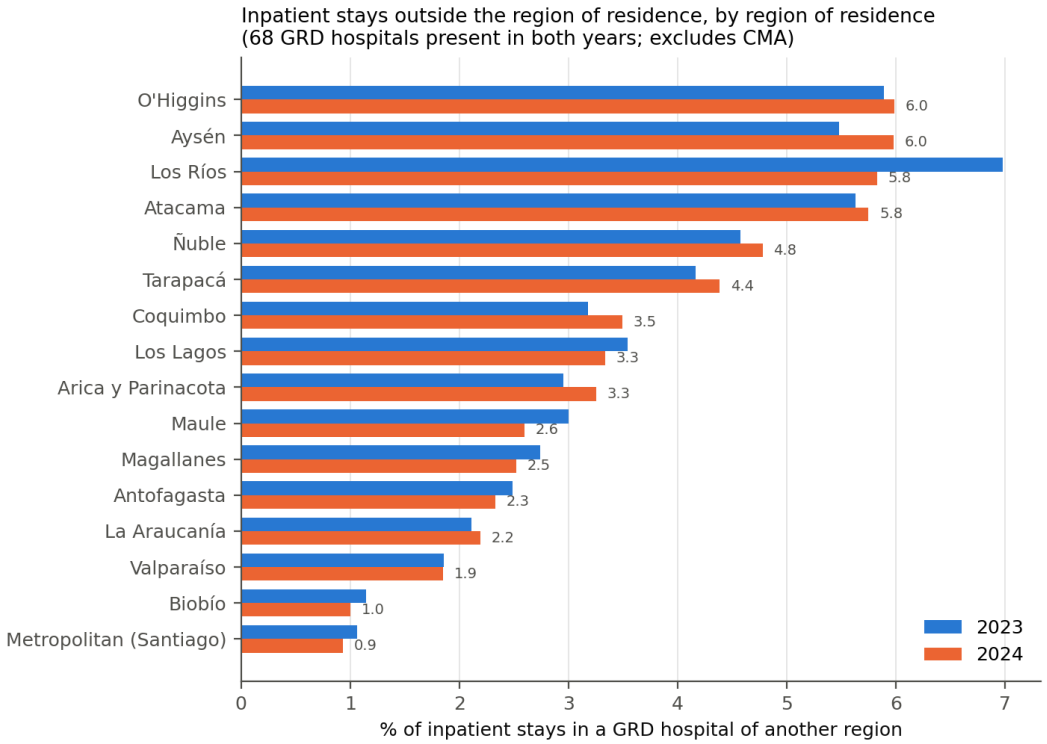


Figure 1. Share of inpatient stays in a GRD hospital of another region, by region of residence (68 common hospitals). Total: 2.4% in 2023 and 2.3% in 2024.

Table 3. Ten most frequent inter-regional flows.

Flow (residence → hospital)	Inpatient stays 2023+2024
O'Higgins → Metropolitan (Santiago)	3,364
Valparaíso → Metropolitan (Santiago)	2,473
Ñuble → Biobío	1,987
Maule → Metropolitan (Santiago)	1,852
Metropolitan (Santiago) → Valparaíso	1,510
Coquimbo → Metropolitan (Santiago)	1,420
Los Ríos → Los Lagos	1,293
Metropolitan (Santiago) → O'Higgins	1,095
Biobío → Metropolitan (Santiago)	1,076
Los Lagos → Los Ríos	973

Interpretation. Leaving the region is uncommon on average (≈ 2.3%), but it is about three to six times more frequent in O'Higgins, Aysén, Los Ríos and Atacama than in the Metropolitan region or Biobío. The Metropolitan region is the main destination, although flows between neighbouring regions also appear (Ñuble → Biobío, Los Ríos ↔ Los Lagos). Leaving the region rises with complexity: it is 4.1% in the highest GRD-weight quartile against 1.2% in the lowest, and 4.7% for elective admissions against 1.9% for emergency admissions. The pattern fits referral of complex, elective cases to reference centres. **Limitation.** The denominator is inpatient stays in GRD hospitals, not the population. A resident treated outside the GRD network is not observed, so the indicator does not measure access or unmet need. Flows from the Metropolitan region to Valparaíso or O'Higgins may reflect temporary stays (holidays) rather than referral.

3.5 Candidate B: first observations

Operational definitions (ours). LOS = discharge − admission, in days (inpatient stays only). Upper outlier = LOS > Q3 + 1.5·IQR within the same GRD code. Expected LOS = mean of the non-outlier episodes of that GRD across all GRD hospitals in 2023-2024. Hospital IEMA = Σ observed LOS / Σ expected LOS (indirect standardisation over non-outliers). No join is needed. 47 episodes with a GRD code outside the dictionary are excluded.

Year	N	Mean	P25	P50	P75	P90	P99	Max	% outlier	% bed-days

										in outliers
2023	846,013	7.10	2	3	7	16	58	696	6.7%	26.2%
2024	873,705	7.05	2	3	7	16	57	661	6.3%	25.4%

Table 4. Length of stay (days) and upper outliers by discharge year. N = analysable inpatient stays.

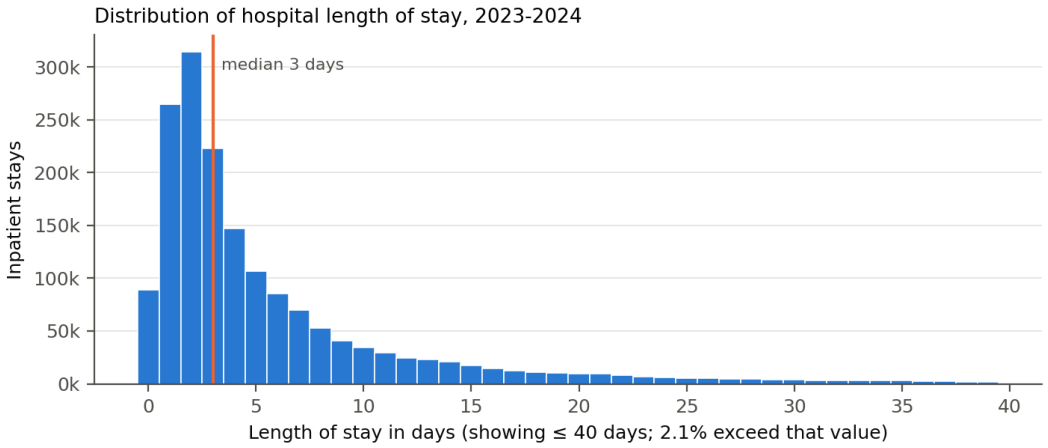


Figure 2. Distribution of length of stay (inpatient stays 2023-2024). The right tail is long: 2.1% exceed 40 days.

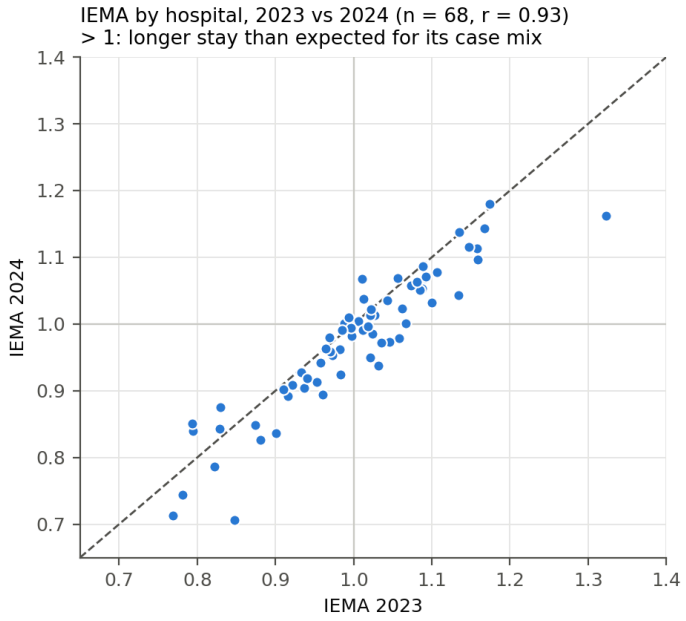


Figure 3. IEMA by hospital, 2023 against 2024 (68 common hospitals). The dashed line is y = x. r = 0.93.

Interpretation. Median LOS is 3 days and the mean ≈ 7. Extreme stays (≈ 6.3% of episodes) account for ≈ 25% of bed-days, with a mean LOS of ≈ 29 days against ≈ 5.6 for the rest. The outlier share is close to the 6.4% of the 2014 norm [6], which is consistent with our operational definition as a first approximation (it does not validate it on its own). In 2024 IEMA ranges from 0.71 (Instituto Traumatológico Dr. Teodoro Gebauer) to 1.18 (Hospital Intercultural de Nueva Imperial), and the ranking is stable between years (r = 0.93). Crude hospital LOS correlates with mean GRD weight (r ≈ 0.53), whereas IEMA hardly does (r ≈ −0.07), which indicates that the case-mix adjustment removes the complexity effect, as intended. **Limitation.** The benchmark is internal, not an official norm. 33 of 790 GRDs have fewer than 30 cases, so their outlier threshold is unstable. A high IEMA may come from complexity not captured by GRD (national referral institutes), internal transfers or waiting for a social discharge, and the files cannot separate these causes.

4. Initial framing with the seven steps (Class 03)

Step	Candidate A	Candidate B
Decision	Where to strengthen regional capacity and how to size referral	Which hospitals and GRDs to target for stay management
Success criteria	A measurable, stable regional gap attributable to complex cases	Excess bed-days identified and a signal stable between years
Business questions	Who leaves, where to and with which cases	Who exceeds the expected stay and how much outliers contribute
Business process	Referral: admission → stay in another region → discharge or return	Admission → stay (with internal transfers) → discharge
Entities / events / states	Patient (residence), hospital (region), episode; event: admission	Hospital, GRD, episode; states: normal or outlier stay
Data representation	Municipality of residence (text), COD_HOSPITAL, dates, GRD	Dates, IR-GRD code, severity, discharge type
Measurement / gap	% outside region; **gap:** no population, no non-GRD hospitals	IEMA and % bed-days in outliers; **gap:** no official norm, no causes

5. Comparison of the alternatives

Dimension	Candidate A · Outside region	Candidate B · Adjusted stay
Relevance supported by sources	Specialist concentration in the Metropolitan region [4]; GRD network [3]; Axis 7 [5] without a verifiable target	GRD used to compare adjusted LOS; 2014 norm [6]; FONASA [3]; Axis 7 [5]
Fit between question and inspected data	Observes leaving the region *within* the GRD network; cannot observe population or external providers	Directly observes LOS and case mix for every episode in the GRD network
Analytical opportunity for C1	Origin-destination matrix by GRD or specialty; relation to regional capacity	Breakdown of excess by GRD, specialty, discharge type and transfers; monthly trend
Main limitation and its consequence	Non-population denominator: cannot speak about access or equity	Internal benchmark: IEMA is relative to the network, not absolute efficiency
Feasible scope and future ML / dashboard potential	Possible ML task: classify the probability of an out-of-region admission. Dashboard for Health Services: flows and % by region. To check: which variables are known at admission	Possible ML task: predict prolonged stay (outlier) from admission variables. Dashboard for bed management: IEMA with intervals and % outlier bed-days. To check: information leakage, as the GRD is assigned at discharge

Provisional preference: B. Its data answer the question without external sources, the signal is stable ($r \approx 0.93$) and it leads naturally into C1: the C1 EDA can characterise prolonged stays, and later stages could consider predicting them and a bed-management dashboard. A remains relevant, but its interpretation depends on a denominator we do not have. **What could change the decision:** (i) if territorial equity is considered a priority for the observatory and DEIS discharges from all facilities or INE population can be added within the C1 scope; (ii) if an IEMA with an internal benchmark is judged insufficient without the official MINSAL/FONASA norm.

6. Questions for feedback

- For C1, is an expected LOS computed internally (mean of non-outliers per GRD within the network) acceptable, or should we look for the official LOS norm per GRD? Do you know of a public source?
- In B, should IEMA be stratified by facility complexity, or should national referral institutes (e.g. Asenjo Neurosurgery, Traumatology Institute) be excluded, since GRD may not capture their case mix?
- For a future prediction of prolonged stay, which variables do you consider legitimately available at admission? The main diagnosis and the GRD are coded at discharge.
- In A, does the C1 scope justify adding DEIS discharges to correct the denominator, or is it better to narrow the question to «referral within the GRD network»?
- Does our reading of the Five Cs match the Class 03 definition?

7. Project continuity (ML and dashboard)

A. Possible task: binary classification (out-of-region admission yes/no) to anticipate referral demand by region and case type. Data: age, sex, municipality, admission type, origin and main diagnosis. To check: whether the signal is dominated by region of residence (in which case a descriptive table is enough and ML adds no value) and class imbalance (≈ 2%). Dashboard: Health Services and Undersecretariat of Healthcare Networks; origin-destination matrix, % outside region by region and GRD, monthly trend.

B. Possible task: classify at admission the risk of a prolonged (outlier) stay, to prioritise discharge management. Data: age, admission type, origin, admitting unit, specialty and insurance category. To check: information leakage (GRD, severity and diagnosis are known at discharge) and stability between years. As required at this stage, no algorithm, validation protocol or accuracy is proposed. Dashboard: hospital directors and bed managers; IEMA with intervals, % bed-days in outliers by GRD and unit, and comparison with hospitals of similar complexity.

Process record

Decisions and checks

Date / stage	Decision	Alternative	Reason and evidence	Check and result	Location
24-09 Sources	Download GRD from the Data Catalogue of the FONASA dashboard	bases-grd path in the course script	The bases-grd page redirects to a 404; the catalogue publishes URLs and extraction dates	URLs read from the catalogue; ZIPs intact; SHA-256 recorded	README; report §3.1
25-09 Extraction	Extract 23 columns + 4 derived; drop patient ID and date of birth	Work with all 129 fields	Limited memory (3 GB) and personal-data minimisation	Extract rows = original rows (1,039,587 / 1,085,813)	analysis/01_extraer_grd.py
25-09 Extraction	Parse each date column with its own format	One format per file	The first extraction left age empty in 2023 (birth YYYY-MM-DD, admission DD-MM-YYYY)	Fixed and re-run: age missing in 0.001%	01_extraer_grd.py; notebook §2
25-09 A	Region of residence from municipality	Modal region per province (reference)	Avoids circularity; uses a patient attribute	No region: 12 episodes	notebook §4
25-09 A and B	Exclude CMA; compare years on 68 common hospitals	All episodes and hospitals	CMA has no stay; 4 hospitals join in 2024	Table 1 and coverage	notebook §1
25-09 B	Internal benchmark per GRD and Q3+1.5·IQR outlier	Official MINSAL norm	No downloadable public norm found	% outlier 6.3-6.7% vs 6.4% in 2014 norm [6]	notebook §5
25-09 Language	Translate all files into English	Keep Spanish text	The course is taught in English	Notebook re-run; figures regenerated with English labels	all files

Instructor feedback and response

Feedback is pending and the provisional decision is to develop B in C1.

Individual contributions

Member	Concrete contribution	Evidence
Matías López	create and develop the file 01_extraer_grd.py	01_extraer_grd.py
	error found (First extraction with empty 2023 age) and fixed (date parser per column)	01_extraer_grd.py
	Contribute to the creation of this report	This report

Cristóbal Ramírez	literature search (Sources [3]-[6] read in their web or PDF version) Contribute to the creation of this report create and develop the file 02_inspeccion_GRD_W1.ipynb	This report This report 02_inspeccion_GRD_W1.ipynb

AI use and verification

Tool and purpose	Relevant excerpt	Accepted / modified / rejected	Verification
Claude: review repositories, locate official URLs, review and help correct minor details on the extraction script, the notebook and this report	Prompt: «use that repo and review the report and the notebook following the instructions for W1 [...] using 2023 and 2024 Fonasa GRD data [...]»	The reference's territorial focus (A) was accepted; B was proposed by the AI. Three assumptions of the reference were rejected (section 3.3)	Every figure in the report is read from data/processed/, generated by the executed notebook; URLs were checked against the FONASA catalogue

Individual reflection

Matías:

Contribution: I built the age-parsing step in the extraction script and found that the first run left `AGE_YEARS` empty for a large share of the 2023 file, traced it to birth dates stored as YYYY-MM-DD while admission dates in the same file used DD-MM-YYYY, and fixed the parser to read each date field with its own explicit format before re-running the extraction.

Learning: I learned that "the same file" doesn't guarantee "the same date convention" two columns in one dataset can silently use different formats, and a parser that assumes one global format will fail quietly (empty values, not an error) rather than crash, so the output has to be checked, not just executed.

unresolved uncertainty: I don't know whether other date-like or numeric fields in the 2023 extract have the same kind of hidden format inconsistency that we haven't specifically tested for, since we only caught this one because age happened to be a field we were already inspecting.

Cristóbal:

Contribution*:* I wrote the notebook section that builds the region-of-residence join mapping each patient's municipality to a region through the facilities table and ran the Five Cs quality checks across the combined 2,125,400 episodes, which surfaced the schema differences between the 2023 and 2024 files (encoding, date format, identifier column).

Learning*:* I learned that a join that looks straightforward (municipality → region) can hide real ambiguity: two municipalities in the source table were listed under two different regions, and resolving that by mode is a judgment call that has to be documented, not just applied silently.

unresolved uncertainty*:* I'm not fully confident in the municipality-to-region mapping, since it relies on the current version of the facilities file rather than the version that would have matched each discharge year; a small share of episodes could be tagged to a region that wasn't accurate at the time of the actual admission.

References

[1] FONASA, Departamento Modelos de Financiamiento and Departamento Estudios y Estadística. «Grupos Relacionados por el Diagnóstico (GRD)», Tableau Public dashboard, Data Catalogue tab (data to December 2024). https://public.tableau.com/views/PropuestaTableroGRD/PropuestaTableroGRD. Files: https://www.fonasa.cl/assets/datos-abiertos/Financiamiento_y_Gasto_en_Salud/GRD_PUBLICO_2023.zip (extracted 04-09-2024) and https://adjuntos.fonasa.gob.cl/assets/datos-abiertos/Financiamiento_y_Gasto_en_Salud/GRD_PUBLICO_2024.zip (extracted 16-05-2025). Retrieved 25-09-2026.

[2] FONASA. «Tablas maestras bases GRD» (xlsx). https://www.fonasa.cl/assets/datos-abiertos/Financiamiento_y_Gasto_en_Salud/Tablas%20maestras%20bases%20GRD%20.xlsx. Retrieved 25-09-2026.

[3] FONASA (October 2023). Boletín Analysis, Edición Especial GRD. https://adjuntos.fonasa.gob.cl/assets/datos-abiertos/Biblioteca/Analysis_octubre2023_Fonasa.pdf

[4] Departamento de Estudios y Desarrollo (July 2023). Caracterización de las especialidades médicas en Chile en 2022. https://www.superdesalud.gob.cl/documentacion/666/articles-26547_recurso_1.pdf

[5] Ministerio de Salud de Chile (2022). Estrategia Nacional de Salud para los Objetivos Sanitarios al 2030. https://www.minsal.cl/wp-content/uploads/2022/03/Estrategia-Nacional-de-Salud-2022-MINSAL-V8.pdf

[6] Zapata, M. (2018). Importancia del sistema GRD para alcanzar la eficiencia hospitalaria. Revista Médica Clínica Las Condes, 29(3), 347-352. https://doi.org/10.1016/j.rmclc.2018.04.010

[7] Course IIB423T-1. Repository Mxtiilxz/Workshop (commit af9d0d7): W1 brief, FUENTES inventory and download script. https://github.com/Mxtiilxz/Workshop

[8] Garrido, L. and Ramírez, C. Repository thenoeyedgirl/proyecto-GRD-Fonasa (commit e87ab54): notebook Analisis_Grupo_5_v2.ipynb and BDDs/establecimientos.csv (MINSAL, current health facilities, https://datos.gob.cl/dataset/establecimientos-de-salud-vigentes; the file's version date is not documented in the repository). https://github.com/thenoeyedgirl/proyecto-GRD-Fonasa